

Nathalia Aparecida Dutra Pires

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EDUCATION

PhD in Astronomy 05/2024 - Present

Project title: Binary systems of compact objects in the Galaxy and the resulting mass distributions;

Fellowship: FAPESP-DD I (grant number 2025/01653-5);

Institute of Astronomy, Geophysics and Atmospheric Sciences of the University of São Paulo.

Bachelor's degree in Physics 01/2020 - 12/2023

Scientific Initiation: Dark Energy influence in the Large Scale Structures formation in the universe;

Fellowship: CNPq (grant number 125181/2022-1)

Institute of Physics of the University of São Paulo.

Technical Course Integrated to Secondary Education 01/2017 - 12/2019

Technical Course: Chemistry;

ETEC Júlio de Mesquita.

PUBLICATIONS

· Scientific articles

de Avellar, M., Horvath, J., **Pires, N. D.**, Rocha, L., de Sá, L., Martins, B., Barão, L., Rodrigues, L., Sampaio, G., Takahashi, R. The mass of the neutron star in 4U 1820-30 revisited. JHEAp, v. 54, 2026;

Benvenuto, O. G. ; de Vito, M. A. ; Echeveste, M. ; Novarino, M. L. ; **Pires, N. D.** ; de Sá, L. M. ; Horvath, J. E. Formation and nature of “Huntsman” binary pulsars. A& A, v. 698, p. L5, 2025;

Rocha, L., **Pires, N. D.**, de Sá, L., Martins, B., Barão, L., Sampaio, G., Rodrigues, D., Takahashi, R., de Avellar, M., Horvath, J., Bernardo, A. The Galactic Black Hole mass distribution (submitted);

Benvenuto, O., **Pires, N.**, Horvath, J. The effects of the irradiation feedback on the “Huntsman” binary systems. (in preparation);

Pires, N., de Sá, L., Gossage, S., Horvath, J. Reproduction of Spider Systems parameters using different Magnetic Braking prescriptions. (in preparation);

Pires, N., de Sá, L., Sampaio, G., Rodrigues, D., Martins, B., Horvath, J. Compact object mergers in the Lower Mass Gap across cosmic time. (in preparation);

Pires, N., de Sá, L., Sampaio, G., Rodrigues, D., Martins, B., Barão, L., Takahashi, R., Horvath, J. Are the objects in the Lower Mass Gap actually in the Gap? (in preparation).

· **Proceedings**

Rocha, L. R., **Pires, N. D.**, de Sá, L., Martins, B., Sampaio, G., Rodrigues, D., Barão, L., Takahashi, R., de Avellar, M., Horvath, J., Bernardo, A. Galactic Black Holes masses. (submitted to the *Astronomische Nachrichten*).

Horvath, J. E. ; de Sá, L. M. ; Rocha, L. S. ; **Pires, N. D.** ; Miwa, R. ; Rodrigues, D. ; Barão, L. G. ; Chinen, G. Y. Investigating the Evolution of Compact Object Populations. *Astronomische Nachrichten*, v. 1, p. 1, 2025.

· **Books**

Horvath, J. ; Rocha, L. ; de Sá, L. ; **Pires, N. A. D.** ; Avellar, M. ; Barão, L. ; Rodrigues, D. ; Takahashi, R. ; Gusmão, L. ; Chenin, G. . O Bê-a-bá da Astrofísica. 1. ed. Livraria da Física, 2025.

EXPERIENCE

Institute of Astronomy, Geophysics and Atmospheric Sciences of the University of São Paulo

· *PhD in Astronomy* 05/2024 – Present

- Project title: Binary systems of compact objects in the Galaxy and the resulting mass distributions;
- Advisor: Prof. Dr. Jorge E. Horvath;
- Fellowship: FAPESP.

· *Scientific Initiation* 06/2021 – 02/2022

- Project title: Analysis of the X-Ray emission in different galaxies;
- Advisor: Prof. Dr. Gastão César B. Lima Neto;
- No fellowship.

Institute of Physics of the University of São Paulo

· *Scientific Initiation* 03 – 09/2023

- Project title: Dark Energy influence in the Large Scale Structures formation in the universe;
- Advisor: Prof. Dr. Luis Raul W. Abramo;
- Fellowship: CNPq (grant number: 125181/2022-1).

· *Teaching Assistant*

- Foundations in Astronomy 02/2026 - present;
- Optics 02 - 07/2025;
- Physics I 08 - 12/2024;
- Physics I 08 - 12/2023;
- Electricity and Magnetism II 03 – 06/2022;
- Physics 2 08 - 12/2021.

Federal University of the ABC (UFABC)

· *Scientific Initiation* 08/2018– 07/2019

- Project title: Natural radioactivity measurement from different types of edible salts;
- Advisor: Prof. Dr. Felipe Chen Abrego;
- Fellowship: CNPq/PIBIC-Jr (grant number: 117440/2018-3).

PARTICIPATION IN EVENTS

XLVIII Reunião Anual da SAB Hotel Glória, Caxambu, Minas Gerais - Brazil.	10/2025
1st POSYDON School The SkAI Institute, SKAI, United States of America.	09/2025
MESA Summer School Katholieke Universiteit Leuven, KU Leuven, Belgium	06/2025
XLVII Reunião Anual da SAB Hotel Majestic, Águas de Lindoia, São Paulo - Brazil.	09/2024
Artificial Intelligence in Astronomy 2023 (AIA 2023) UNESP	09/2023
New Particle Accelerators and the Future of Particle Physics ICTP - SAIJR	07/ 2023
Journeys into Theoretical Physics Perimeter Institute – ICTP-SAIJR	07/2023
The Hawking Radiation Entropy ICTP - SAIJR	06/2022
Minicourse in Phase Transitions and Gravitational Waves ICTP - SAIJR	02 – 03/ 2023
Program on New Directions in Particle Physics ICTP - SAIJR	09/2022
Journeys into Theoretical Physics Perimeter Institute – ICTP-SAIJR	08/2022
Summer Course 2022 Institute of Physics of the University of São Paulo	03/2022
Introduction to the Mathematical Gauge Theory Jayme Tiomno School of Theoretical Physics	08/2021
Introduction to Superstring Theory Jayme Tiomno School of Theoretical Physics	08/2021
Cosmology Jayme Tiomno School of Theoretical Physics	08/2021
Real Analysis National Institute of Pure and Applied Mathematics (IMPA)	01 – 03/2021

TALKS

RASAB 2025 <i>Poster</i>	10/2025
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Presentation of the Poster: "*Formation and Nature of Huntsman Binary Systems*".

RASAB 2024

09/2024

Poster

Presentation of the Poster: "*The Galactic BH mass distribution*".

SIICUSP (International Symposium of Scientific Initiation of the University of São Paulo)

10/2023

Symposium

Presentation of the Scientific Initiation project: "*Dark Energy and the formation of Large Scale Structures in the universe*".

V Jayme Tiomno School of Theoretical Physics

09/2023

Presentation

Presentation about the perspectives of the use of Gravitational Waves as standard-sirens in Cosmology.

IX Meeting of Scientific Initiation of the UFABC

11/2019

Symposium

Presentation of the Scientific Initiation project: "*Natural radioactivity measurement from different types of edible salts*".

LANGUAGES

Portuguese (native), English (fluent), Spanish (advanced), French (basics).